

PANAVERSITY AI-101

GAPS & The Agent Factory

General Agent Problem Solving

The Evolution of Intelligent Systems

From knowledge retrieval to autonomous problem-solving architectures.

Beyond Knowledge Retrieval

The Intelligence Bottleneck

Traditional AI focus was on **Knowledge**. Modern AI focus is shifting to **Expertise & Execution**.

- ✓ **Old:** Chatbots summarizing data.
- ✓ **New:** Agents solving multi-step tasks.
- ✓ **Goal:** Delivering outcomes, not just text.

[PROCESS]

Input: Problem Goal

Agent: Reasoning Pattern

Output: Verified Solution

| What is GAPS?



General

Not confined to specific tasks; able to reason across diverse domains using universal logic.



Agent

An autonomous entity that perceives environment and acts upon it to achieve goals.



Problem Solving

The core objective. Moving from input to a verified, executed solution.



Manufacturing Intelligence

Designing the systems that build the agents.

| The Agent Factory

The **Agent Factory** is an assembly line for AI intelligence. It doesn't just use AI; it manufactures specific behaviors.

 **Standardization:** Creating repeatable agent patterns.

 **Componentization:** Separating logic from tools.

 **Validation:** Continuous testing of agent reliability.



Agent Workers: The Workforce



Distributed Execution

Agent Workers are the specialized entities that execute tasks within the factory framework.

-  **Orchestration:** Workers report to managers.
-  **Tool-Bound:** Each worker has specific toolsets.
-  **Focus:** Solving discrete sub-problems.

| The 7 Principles of Agentic Systems



1. Autonomy

Independence in decision-making within defined guardrails.



2. Goal Orientation

Prioritizing the objective over the specific instruction.



3. Adaptability

Modifying behavior based on real-time environment feedback.

| The 7 Principles Continued



4. Proactivity

Taking initiative rather than just reacting to prompts.



5. Social Ability

Collaborating with other agents and humans seamlessly.



6. Persistence

Maintaining state and goals over long-running operations.



7. Accountability

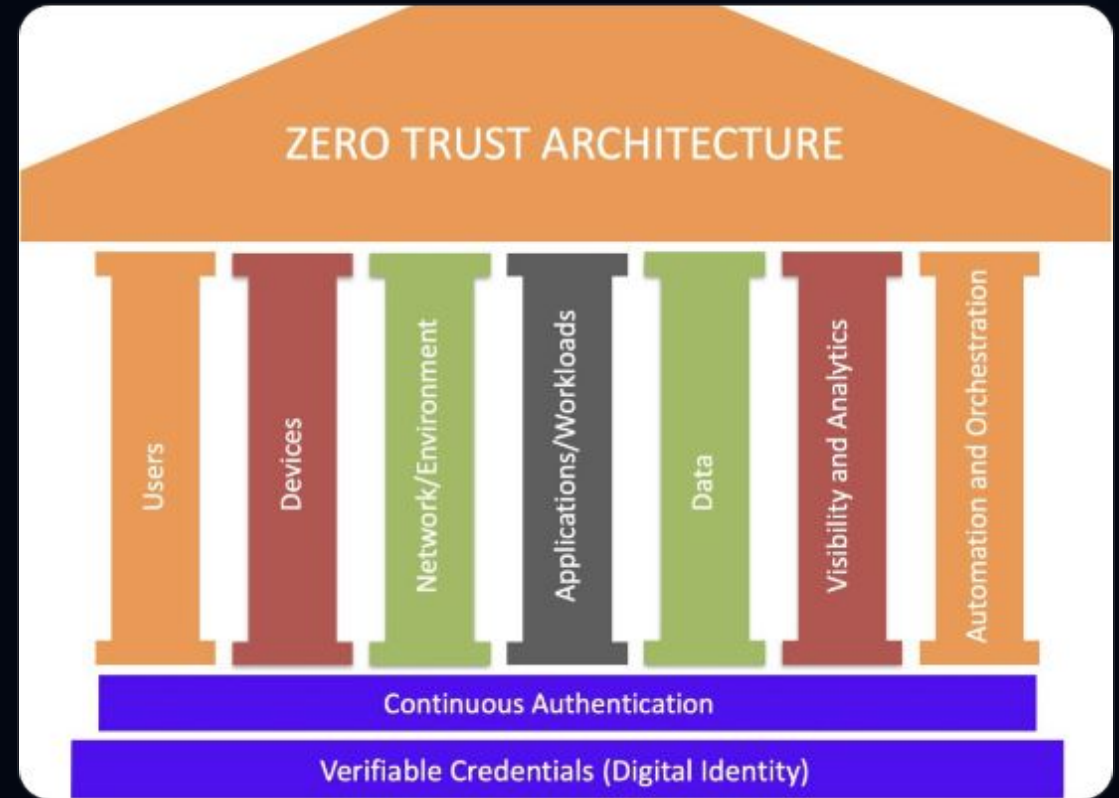
Traceability of actions and adherence to ethical constraints.

Expertise > Raw Intelligence

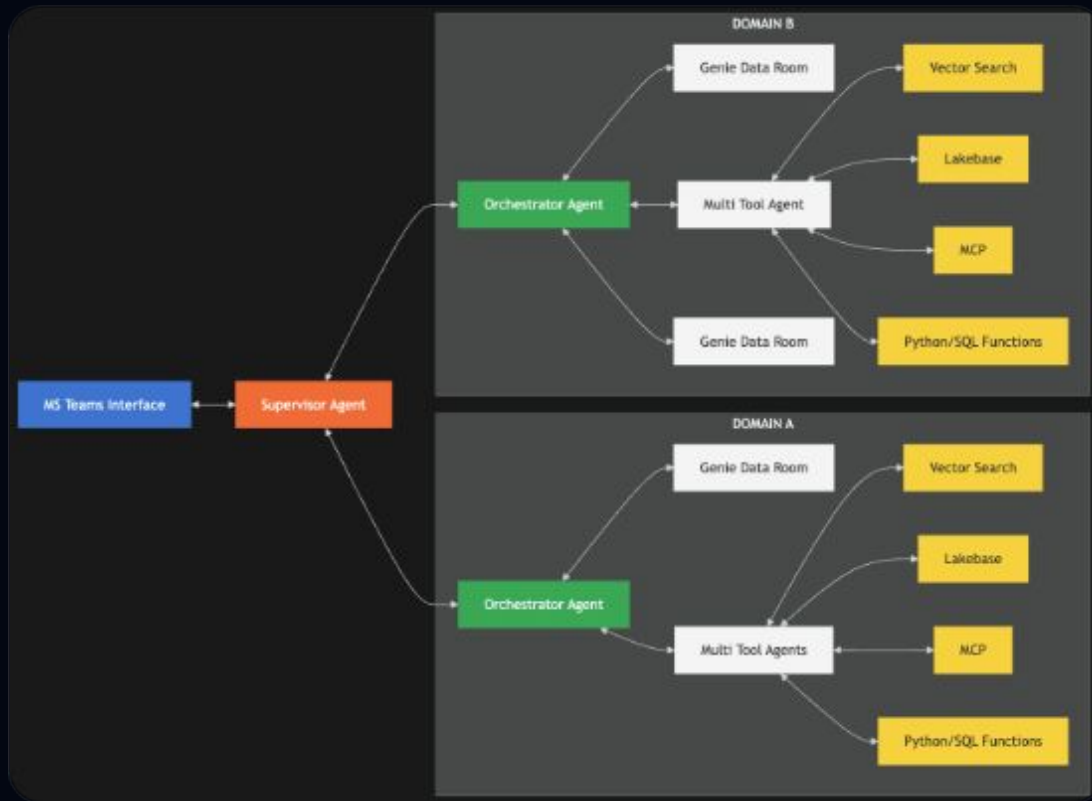
Intelligence is a processor; **Expertise is the application.**

We don't need smarter models; we need models that can access the right skills at the right time.

-  **Skill Folders:** Modular capabilities.
-  **MCP Protocol:** Linking tools to models.
-  **Context:** Using the model as a reasoner.



Multi-Agent Collaboration



Role

Primary Responsibility

Orchestrator

Planning and Goal Decomposition

Worker

Task Execution and Tool Usage

Reviewer

Quality Assurance and Verification

| Implementation Roadmap



Discovery

Identifying high-impact problem domains.



Design

Architecting the GAPS workflow.



Factory Build

Setting up the assembly line & MCP.



Deployment

Scaling agents with verified expertise.

Key Takeaway

"The future of AI is not in larger models, but in **architecting factories** that deliver autonomous, verified problem-solving through general agents."

Questions?

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Revolution

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| Image Sources



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